Estimation of Housing Rental Prices in the USA

Group E

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Abstract

This study examines the key factors impacting rental prices for housing in the United States. This work utilizes the UCI " Apartment for rent Classified" dataset. The goal is to create reliable and useful predictive models that can help renters and landlords understand and approximate rent prices for where the market is at currently. In addition, the project incorporates some text data extraction using large language models (LLMs) to develop structured variables from unstructured advertisement descriptions (e.g., amenities, location, and unit features). After significant preprocessing and engagement with engineering features, many supervised machine learning algorithms were examined including: Random forest, XGBoost, Support Vector Machines, and Multinomial Logistic Regression. Random forest achieved the highest predictive accuracy of 89%, indicating that this model is the most reliable for predicting categories of rental price. The analysis of feature importance sufficiently demonstrated that geographic relative to standard geographic coordinates (e.g., longitude, and latitude) are the most impactful variables in determining categories for rental price, followed by the number of bedrooms, and amenities. This research supports decision-making for Veterans and others looking for transparency in areas of the U.S. housing market to promote equitable housing.

1. Introduction

The rental housing market in US is one of the largest in the world. Many Americans rent apartments rather than rent homes. However, the rental prices can vary greatly depending on the location, the type of the house etc. These factors should be understood by renters and providers of the rental housing. Price estimations must be accurate to assist renters in making rational housing decisions. This prevents renters from paying too much and helps landlords arrive at competitive market rates that minimize the rental rate disparity and vacancies, thus, reflecting true market conditions.

This research focuses on the primary factors determining US housing market rent pricing concentrating on UCI "Apartment for Rent Classified" data. The aim of the research is to assist in reasonably accurate pricing of hosing rentals using predictive models and to understand the impact of property and listing variables on rental pricing. This research also address the issues of handling attribute and text data, the practical applicability of the predictive models, and the applicability of the results in the real estate domain.

The results of the research assist members of the housing market in making rational decisions. It demonstrates how research helps in the transparency of the urban rental housing market to make equitable pricing and efficiency urban planning possible.

2. Description of the Question

Objective 1 Dataset Exploration

Exploratory data analysis (EDA) allows for the understanding of the principal elements that determine rental prices.

Objective 2: Information Extraction from Unstructured text

Extract structured information (rent amount, bedrooms/bathrooms, unit type, location, amenities, lease terms) from unstructured listing text using LLMs.

Objective 3: Predictive Modeling

Develop and evaluate machine learning models capable of accurately estimating rental prices.

Objective 4: Explainable AI (XAI) Integration

Apply SHAP and other XAI techniques to provide predictability, interpretability, and reliability for the provided model outcomes.

Objective 5: Web Application Development

Create a real-time web application that predicts and rationalizes rental prices.

3. Information Extraction from Unstructured Text using Large Language Models.

The process of converting raw advertisement text into structured data involved several key stages, from initial text processing to the final creation of a structured dataset. The entire workflow was executed using Kaggle notebooks, taking advantage of their GPU resources to handle the computationally intensive tasks.

3.1. Initial Text Analysis and Key Phrase Extraction

The first step was to break down the unstructured 'body' of the rental advertisements into more manageable and meaningful chunks. Instead of simple sentence splitting, a more advanced approach was taken using the google/gemma-3-4b-it model. This model was prompted to identify and extract key phrases from the advertisement text. This method helps to preserve the context and group related information together, such as a full address or a price range with its corresponding unit types.

3.2. Embedding Generation and Thematic Clustering

Once the key phrases were extracted, the next step was to understand the thematic structure within them. This was achieved by:

1. **Generating Embeddings:** The google/embeddinggemma-300m model was used to convert each key phrase into a high-dimensional vector representation (embedding). These embeddings capture the semantic meaning of the phrases.
2. **Dimensionality Reduction:** The high-dimensional embeddings were then reduced to a lower-dimensional space using Uniform Manifold Approximation and Projection (UMAP). This makes it possible to apply clustering algorithms effectively.
3. **Clustering:** The HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) algorithm was applied to the reduced-dimension embeddings. This grouped semantically similar key phrases together, helping to identify common themes present in the advertisements, such as mentions of specific amenities, location details, or rental terms.

3.3. Information Extraction with a Large Language Model

With a better understanding of the content through clustering, the core information extraction task was performed. The google/gemma-3-4b-it model was again utilized, but this time with a more complex prompt designed for structured data extraction.

A detailed prompt was engineered that included:

- **A clear set of rules** for how to interpret the advertisement text.
- **A specific JSON schema** that the model had to follow for its output.
- **Few-shot examples** to guide the model in producing the correct output format.

This instructed the model to analyze the raw advertisement text and extract information related to rental price, unit details, amenities, and location, formatting it as a JSON object.

3.4. Data Post-processing and Structuring

The JSON objects produced by the LLM were then post-processed to create a final, structured dataset. A custom script was developed to handle the logic for price allocation. When a price range was provided for a range of unit types (e.g., "$800-$1200 for 1-3 bedrooms"), the script interpolated the prices across the different unit types. The processed data from each advertisement was then compiled into a single, clean tabular dataset and saved as a CSV file. Due to the large volume of data, this process was done in chunks, and the resulting CSV files were later merged into a final comprehensive dataset.

4. Description of the Dataset

The information extraction process resulted in a structured dataset containing 12,376 individual rental unit listings. Each row in the dataset represents a specific unit type at a given price, derived from the original advertisement text. The dataset comprises 17 features, which are a mix of categorical and numerical data, describing the amenities, location, and key characteristics of each rental property.

The table below provides a detailed description of each variable in the final dataset.

Variable	Description	Type
Parking	Indicates whether parking is available. Values are provided or not.	Categorical
Flooring	Describes the primary type of flooring in the unit. Values include wood, carpet, tile, or other.	Categorical
Laundry	Specifies if laundry facilities are available. Values are provided or not.	Categorical

Dishwasher	Indicates the presence of a dishwasher. Values are provided or not.	Categorical
Stainless_Appliances	Notes whether the unit includes stainless steel appliances. Values are provided or not.	Categorical
Cable	Specifies if cable service is included with the rent. Values are included or not.	Categorical
Internet	Specifies if internet service is included with the rent. Values are included or not.	Categorical
Outdoor_Spaces	Describes private outdoor spaces. Values are Balcony, Patio, Deck, Combined (if multiple types are listed), or No Outdoor Space.	Categorical
AC	Indicates if the unit has air conditioning. Values are provided or not.	Categorical
Pool	Specifies if the property has a pool. Values are provided or not.	Categorical
Fitness_Facilities	Indicates if the property has on-site fitness facilities. Values are available or not.	Categorical
State	The two-letter postal code for the state where the property is located (e.g., 'TX', 'CA').	Categorical
City	The city where the property is located.	Categorical
Rental_type	The rental payment frequency. The predominant value is Monthly.	Categorical
Price	The monthly rental price in US dollars for the unit.	Quantitative
Unit_Type	A description of the unit size, such as studio, 1 bedroom, 2 bedroom, etc.	Categorical
Bedrooms	The number of bedrooms. Studio apartments are coded as 0.	Categorical

Table 1 Description of the Dataset

4. Pre-Processing

- A total of 481 duplicate records were identified and removed from the dataset to ensure data quality and avoid bias.
- Missing values were found. Since missing rate is less than 0.02% , the corresponding records were dropped.
- Split data into two sets as training (70%) and testing (30%). Descriptive analysis was conducted using the training set.

5. Feature Engineering

- The continuous Price variable was convert into four categorical ranges.

  Price < 975 : Low range (1)

  975 < Price < 1325 : Mid range (2)

  1325 < Price < 1750 : High range (3)

  1750 < Price : Premium or luxurious (4)
- By using City and State variables, two new variables (longitude and latitude) are created. Then drop City and State variables.

6. Important Results of Descriptive Analysis

6.1 Target Variable-Price Category

The bar char in figure 1 shows the distribution of the four price categories. All categories have a fairly balance distribution with count around 2000 each. This indicates that the dataset is well balanced. This is helpful for training the model.

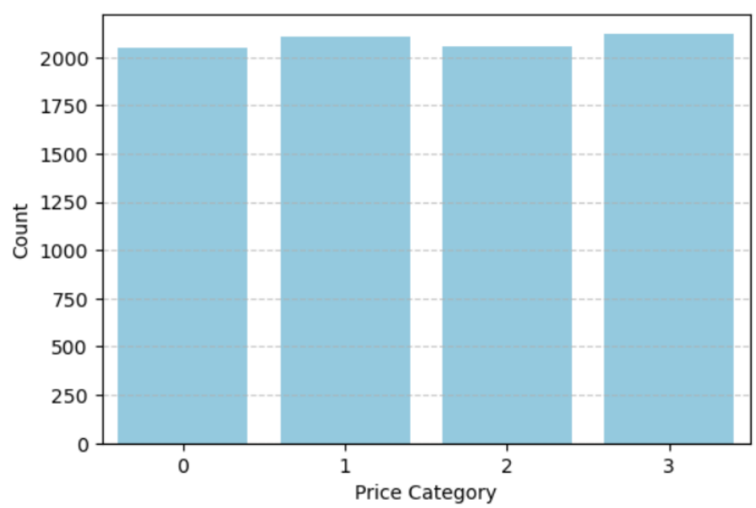


Figure 1 Bar chart of Price category

6.2 Bivariate Analysis

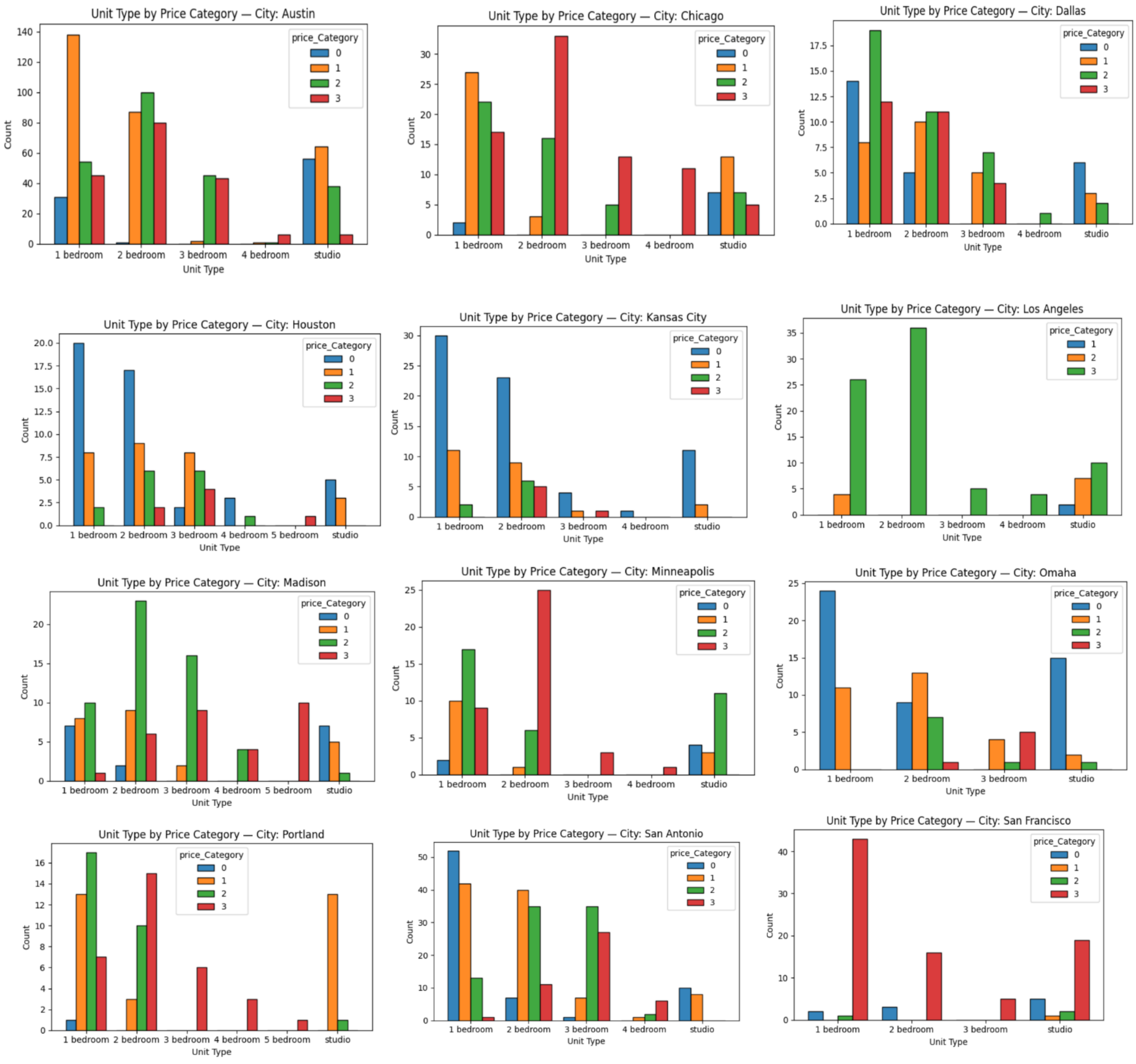


Figure 2 Unit Type by price category in each city

By using figure 2 , we can find how different unit types are priced and sold across the US city. We can see that 1 bedroom units dominate the rental listings in most cities such as Austin, Houston and San Francisco, highlighting their strong demand among urban renters. Most of these units fall within the low to mid range price categories, indicating that the bulk of the available housing is aimed at the affordable and mid market levels, leaving the higher end market aside.

Houses in Austin and San Antonio stand out in having a more balanced price distribution and more ease across the levels of cities. This also provides evidence that these cities provide use diversity. Los Angeles and San Francisco in comparison, feature a lot more expensive 1 bedroom units highlighting the small space high rent phenomena. This shows that people are more likely to pay higher than average market prices for these units.

Apartments with more than 3 or 4 bedrooms are predominately in the mid and higher price category, especially in Austin, Kansas city and Minneapolis. This shows that they are willing to pay the price for these larger units. In comparison, studio apartments are limited and more so in the lower and mid tiers of pricing.

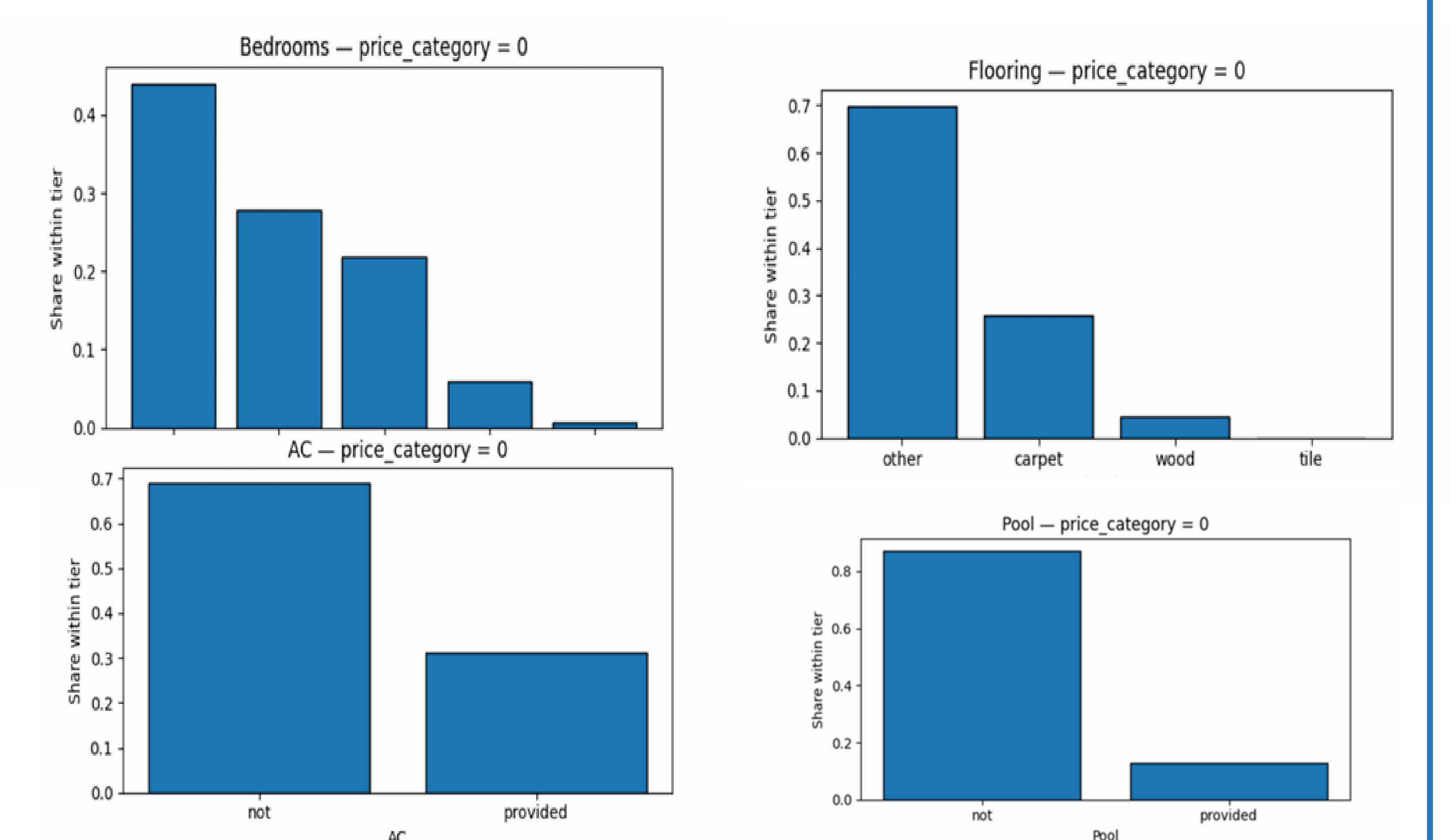


Figure 3 Low range category

Mostly very small homes—studios and 1BR—with a higher share of rooms/sublets than whole units. Amenity levels are low (laundry, dishwasher, AC are uncommon; gym/pool/outdoor access is rare) and finishes are basic. These listings cluster in lower-cost cities or fringe areas.

Figure 4 Mid range category

Moves into entry-level whole units, still 1BR-heavy with some compact 2BR. Essentials like AC and in-unit laundry appear more regularly, but building facilities remain limited or patchy. Finishes begin to improve and this tier is spread across many cities.

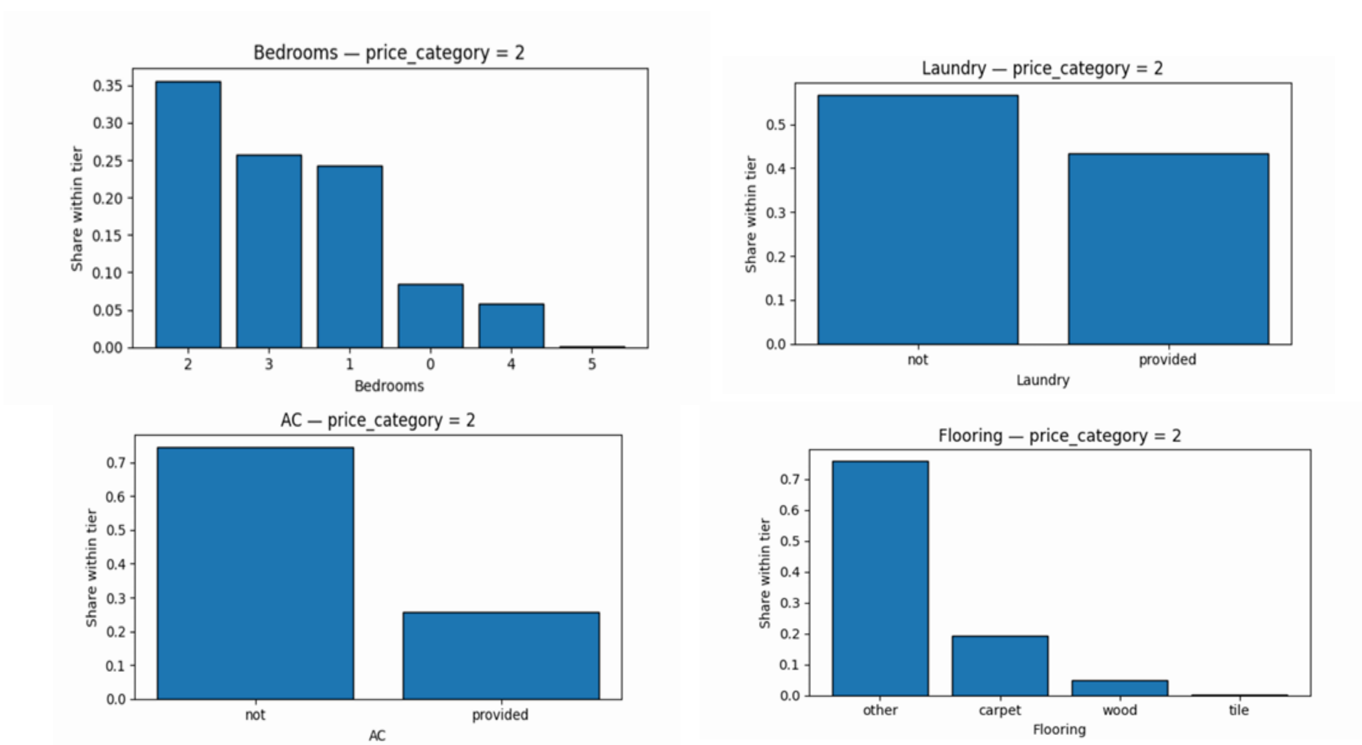


Figure 5 High range category

Size steps up—2BRs are common and unit types skew to standard apartments/condos. Core amenities (laundry, AC, dishwasher) are frequent, and facilities (gym, pool, outdoor/common areas, often parking) are regularly included. Finishes are clearly better, and listings concentrate more in mid- to high-cost cities.

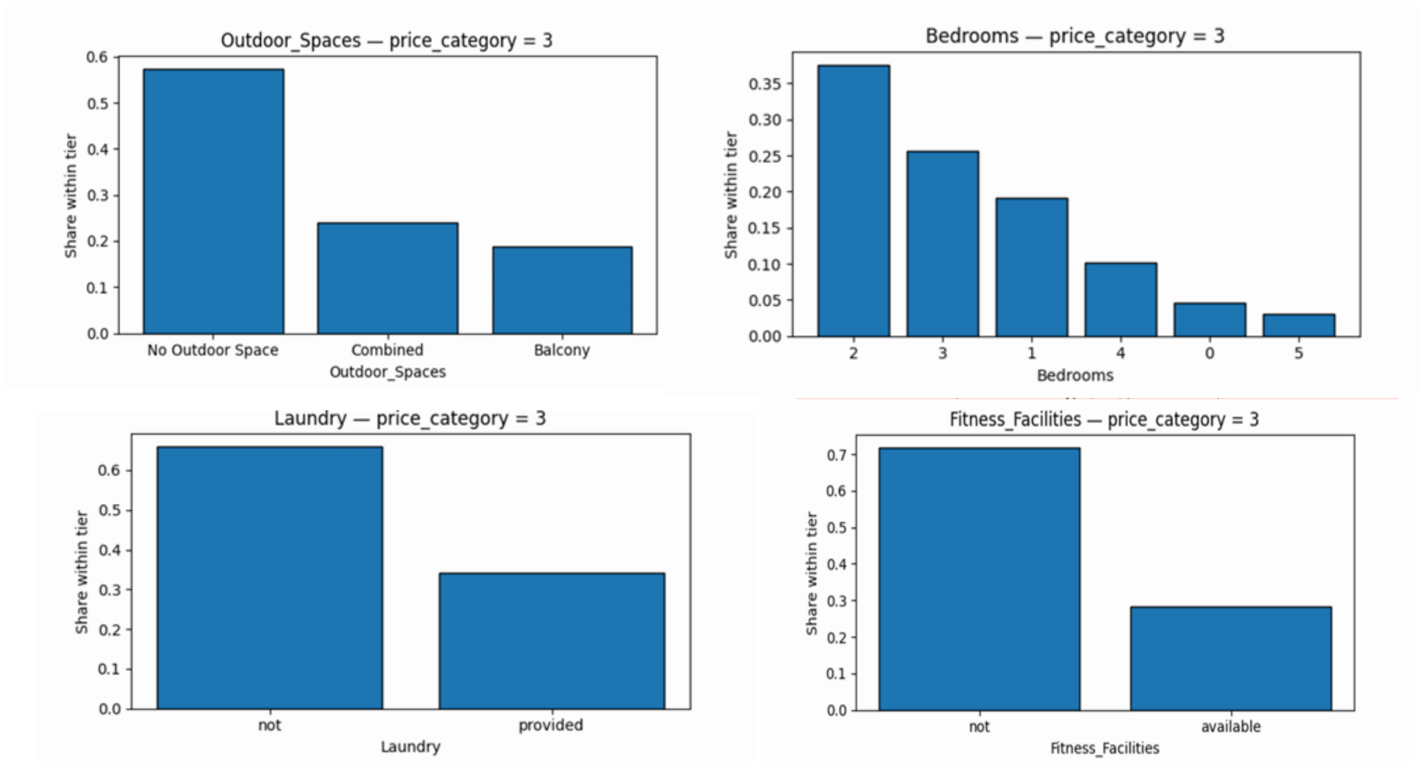


Figure 6 premium or luxurious category

Largest floorplans (2–3BR+) and higher-end types (condos/lofts) dominate. A near-standard full bundle—laundry, dishwasher, central AC, gym, pool, outdoor spaces, and parking—pairs with premium flooring and upgraded interiors. These cluster in the priciest locations and represent the top of the market.

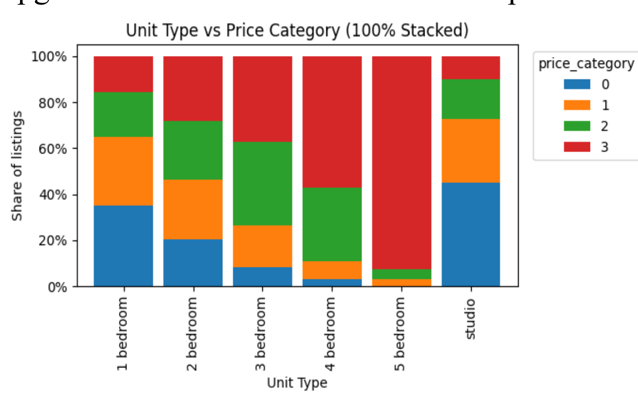


Figure 7 Unit Type vs. Price Tier

From the chart shows in figure 5, **studios** are mostly **low range (0)** with some **mid (1)** and very little **premium/luxurious (3)**. **1-bedrooms** still lean **low (0)** but have more **mid (1)** than studios. **2-bedrooms** shift upward—mainly **mid (1)** and **high (2)** with less **low (0)**. **3-bedrooms** are largely **high (2)** and **premium/luxurious (3)**. **4-bedrooms** are overwhelmingly **premium/luxurious (3)** (some **high (2)**), and **5-bedrooms** are almost entirely **premium/luxurious (3)**. Overall: more bedrooms ⇒ higher share of **high (2)** and **premium/luxurious (3)** tiers.

	feature	n_levels	chi2	p_value	cramers_v
12	Unit_Type	6	1353.799925	1.433976e-279	0.233313
13	Bedrooms	6	1353.799925	1.433976e-279	0.233313
1	Flooring	4	236.653402	6.522869e-46	0.097548
11	rental_type	2	166.894786	5.956599e-36	0.141887
2	Laundry	2	117.803212	2.293596e-25	0.119207
8	AC	2	81.083110	1.797596e-17	0.098898
7	Outdoor_Spaces	5	101.411870	2.943742e-16	0.063857
3	Dishwasher	2	53.837433	1.215305e-11	0.080587
5	Cable	2	44.639155	1.103951e-09	0.073381
10	Fitness_Facilities	2	43.756947	1.699686e-09	0.072652
9	Pool	2	36.757306	5.178824e-08	0.066588
0	Parking	2	30.566692	1.048721e-06	0.060722
6	Internet	2	14.898693	1.905301e-03	0.042393
4	Stainless_Appliances	2	5.828388	1.202647e-01	0.026515

Table 2 Chi-square Test and Cramér's V Results

Table 2 shows that Unit Type and Bedrooms have the strongest correlation with the target variable and Rental Type , Laundry and Flooring have moderate affect for target variable. AC, Dishwasher and cable have only a small influence , while Internet and Stainless Appliances have almost no effect on price differences.

States	Prices
Hawaii	3605.30
California	2686.97
Massachusetts	2611.49
Maine	2400.00
DC	2318.85
New Jersey	2273.25
Rhode Island	2192.45

In the Top 10 States,

The values run from about $1,752 to $3,605. The highest is roughly $3,605, followed by about $2,687 and $2,611. The average across these ten numbers is around $2,346 and the median is about $2,296. The overall spread (max–min) is roughly $1,853, but most values cluster in a typical band of $1.8k–$2.8k, with the top value clearly standing above the rest.

New Hampshire	1852.86
New York	1764.01
Washington	1752.22

Table 3 Top 10 Highest Rental States

6.3 Cramér's V Association Analysis

- Most variable pairs show weak to moderate associations (Cramer's V < 0.6), indicating that the categorical variables are relatively independent.
- However there were few strong associations (Cramer's V >= 0.6)
  - `Laundry ↔ Dishwasher : 0.800`
  - `Unit_Type ↔ Bedrooms : 1.000`
- Since there is a perfect association between unit_type and bedrooms, we dropped the unit_type variable.
-

7. Important Results of Advanced Analysis

7.1 Best Model

Table 3 indicates the accuracy of various machine learning models. Random Forest was the most accurate among the tested models(0.89) . The XGBoost model had a fairly close accuracy(0.82). On the other hand, Multinomial Logistic Regression and SVM had considerably lower accuracies.

	Training set	**Testing set**
Multinomial Logistic Regression	0.423	0.434
Random Forest	0.865	0.891
XG Boost	0.765	0.829
SVM	0.378	0.354

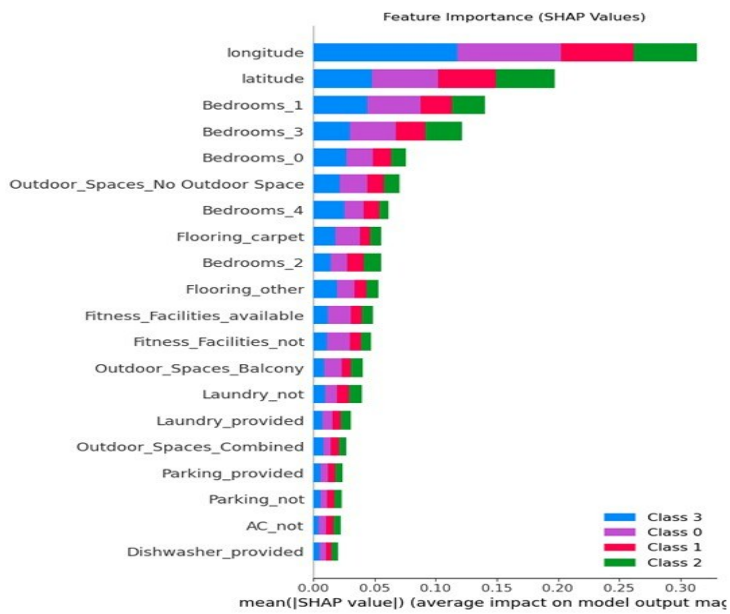
Table 4 Accuracy on the several machine learning models

7.2 Final Model

	Training set	Testing set
Random Forest	0.865	0.891

Table 5 Final model

7.3 Feature Importance Plot



The bar char in figure 6 shows the top 10 feature importances from the random forest model. The most significant features are longitude and latitude, indicating that location is the main factor influencing the target variable's prediction.

The features related to the number of bedrooms show moderate importance, contributing to the prediction but far less than geographic coordinates.

Figure 8 Feature important plot

7.3 SHAP Plot

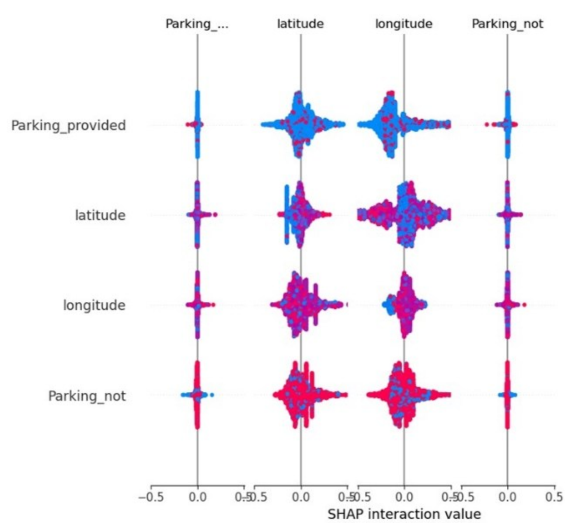


Figure 9 SHAP plot

The SHAP interaction summary plot illustrates how pairs of features jointly influence the model's predictions. The strongest interaction is observed between latitude and longitude, indicating that location has a dominant and interdependent effect on the target variable. This suggests that specific geographic combinations play a crucial role in determining predicted values. Additionally, moderate interaction effects are seen between parking availability (Parking_provided and

Parking_not) and location variables, implying that the influence of parking differs across regions. Overall, the plot highlights that location is the most critical factor, and its impact is further shaped by whether parking facilities are available or not.

8. Discussion and Conclusions

The research effectively examined the intricate relationships between location, attributes of the property, and amenities that affect rental prices in the American housing market. By implementing a structured method of extracting data via large language models, rental listings were transformed from unstructured form into datasets that are useable for our analysis. This allowed us to be more informed about aspects of property characteristics that impact rental value in cities and states.

Exploratory analysis showed that location and size of the property could be characterized as more dominant price drivers as compared with other variables. Cities like Los Angeles and San Francisco have higher average rental prices for one-bedroom listings (and all other listings), which could likely be attributed to the pressures of urban housing and locational quality. Some amenities (laundry, air conditioning, and dishwashers) also showed a somewhat moderate level of impact in differentiating prices, while other amenities (like pools, and gyms) were more likely found in higher-priced property types (i.e., luxury).

The Random Forest model saw the best performance among all of the algorithms at 89% accuracy, and the XGBoost model scored 82% accuracy. The overall performance suggests that an ensemble of models is a stronger approach to capturing the non-linear relationships and interactions that develop between features when predicting a pricing category for rental type. Further analysis of the most important features suggested that geographic positioning (longitude, and latitude) is the most important feature influencing price, which indicates that geographic location is still a strong indicator of rental pricing.

From a practical standpoint, the results will help renters, landlords, and policymakers make data-informed decisions. Renters can better evaluate price equity, landlords can competitively price their properties, and urban planners can track affordability patterns by geographic region. The incorporation of explainable AI methods enhances transparency and trust within the predictive models making them more suitable for real-world use.

In summary, this study highlights that integrating machine learning with advanced natural language processing offers a powerful and reliable framework for analyzing and predicting rental housing prices. Future research could further improve predictive accuracy and policy relevance by incorporating time-sensitive market dynamics, external economic indicators, and richer geospatial data layers. Additionally, during the information extraction stage, large language models (LLMs) could be fine-tuned specifically for rental data extraction, enabling more accurate and scalable processing of diverse textual sources.

9. Appendix including Python code and technical details

https://drive.google.com/drive/folders/1K-Qe_uQX9w4WKTgcPc1wrK7o-gw4lX7n?usp=drive_link